

HILLSDALE, MI, USA

WMO: 725409

Lat: 41.921N

Lon: 84.586W

Elev: 360

StdP: 97.07

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-14.8	-21.9	0.5	-16.5	-18.8	0.7	-13.6	10.5	-4.4	9.2	-3.9	2.3	240	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.7	22.6	30.1	21.9	28.6	21.3	24.8	28.9	23.8	27.8	22.9	26.6	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.1	27.0	22.5	18.0	26.1	22.0	17.4	25.5	77.1	29.2	73.2	27.7	69.7	26.9	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.3	DB	-21.4	33.3	3.2	1.7	-23.7	34.5	-25.6	35.5	-27.4	36.4	-29.7	37.6	(4)
(5)			WB	-21.7	26.1	3.2	1.2	-24.0	26.9	-25.8	27.6	-27.6	28.3	-29.9	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-4.5	-3.3	2.3	9.1	15.2	20.1	22.0	20.8	17.6	10.9	4.4	-1.4	(6)	
	DBStd	10.58	6.21	5.99	6.20	5.19	4.77	3.63	3.27	2.90	4.15	5.00	5.31	5.58	(7)	
	HDD10.0	1745	449	373	251	77	9	0	0	0	1	51	180	354	(8)	
	HDD18.3	3578	707	606	498	281	122	23	6	10	63	236	417	611	(9)	
	CDD10.0	1564	1	1	12	49	170	302	372	336	229	79	13	1	(10)	
	CDD18.3	356	0	0	1	3	25	75	119	87	41	6	0	0	(11)	
	CDH23.3	2972	0	0	7	26	226	621	1058	654	338	42	0	0	(12)	
	CDH26.7	827	0	0	1	3	49	183	342	151	92	6	0	0	(13)	
(14) Wind	WSAvg	3.1	3.7	3.7	3.7	3.9	3.1	2.6	2.1	1.9	2.2	2.9	3.4	3.6	(14)	
(15) Precipitation	PrecAvg	895	51	43	68	84	92	97	91	85	82	67	72	64	(15)	
	PrecMax	1104	126	114	136	140	183	208	205	185	163	218	176	130	(16)	
	PrecMin	581	5	7	19	25	36	13	26	14	8	4	20	19	(17)	
	PrecStd	120	32	26	28	29	36	40	42	46	40	40	32	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	14.8	23.6	26.3	30.5	32.5	34.2	32.1	32.0	27.6	21.1	14.8	(19)	
		MCWB	11.2	11.4	17.1	16.2	20.8	22.0	23.6	23.2	21.8	20.0	14.3	13.8	(20)	
	2%	DB	9.2	11.3	18.0	23.0	27.8	31.0	32.1	29.9	29.2	23.9	17.4	11.5	(21)	
		MCWB	7.9	9.6	13.2	15.0	18.9	22.1	23.4	22.4	21.0	17.0	12.7	9.8	(22)	
	5%	DB	6.9	7.5	14.9	21.0	26.2	28.8	30.2	28.0	27.3	21.4	15.0	8.2	(23)	
		MCWB	5.1	5.6	11.1	13.4	18.4	20.7	22.6	21.6	19.9	16.1	11.4	6.7	(24)	
	10%	DB	3.3	4.2	12.1	17.8	23.6	27.2	28.1	27.1	25.1	19.0	12.7	6.0	(25)	
		MCWB	2.0	2.1	8.6	12.0	17.5	20.4	21.7	21.1	19.0	15.1	9.6	4.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.4	17.4	19.4	22.6	25.0	26.6	25.4	24.0	21.5	15.9	13.8	(27)	
		MCDB	12.5	13.1	23.9	24.4	27.0	29.4	30.3	29.8	28.7	26.0	19.2	14.6	(28)	
	2%	WB	8.1	9.2	14.3	17.0	21.2	23.8	25.3	24.1	22.9	19.2	13.9	10.2	(29)	
		MCDB	8.6	11.1	17.4	21.1	25.3	28.0	29.5	27.7	27.0	22.1	16.4	11.2	(30)	
	5%	WB	5.0	5.7	11.9	14.8	19.9	22.7	24.1	23.3	21.7	17.4	12.0	7.1	(31)	
		MCDB	6.6	7.9	14.7	18.8	24.3	26.6	28.2	26.5	25.0	19.7	14.3	8.0	(32)	
	10%	WB	2.4	2.6	8.4	12.8	18.6	21.6	23.1	22.1	20.3	15.5	9.9	4.4	(33)	
		MCDB	3.6	4.4	11.3	17.4	22.4	25.4	26.8	25.4	23.5	18.7	12.8	5.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	9.0	10.7	10.8	10.5	11.0	10.6	11.4	10.0	8.0	6.3	(35)	
		MCDBR	9.1	10.1	12.9	13.9	13.0	12.2	12.0	11.7	13.3	13.0	10.8	8.8	(36)	
	5% WB	MCWBR	7.9	8.4	8.9	7.9	6.7	5.5	5.3	6.0	6.2	7.2	7.7	7.7	(37)	
		MCWBR	8.6	10.0	11.8	12.2	11.2	10.2	10.1	9.6	10.6	10.8	10.1	8.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.342	0.347	0.388	0.416	0.435	0.425	0.424	0.380	0.345	0.314	0.300	(40)	
		taud	2.378	2.220	2.345	2.253	2.217	2.202	2.241	2.238	2.345	2.437	2.475	2.408	(41)	
	Edn at Noon		846	850	898	881	861	843	847	836	851	845	826	811	(42)	
		Edh at Noon	81	111	111	131	140	142	135	132	110	89	74	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.51	2.42	3.54	4.60	5.35	6.01	6.10	5.28	4.30	2.81	1.77	1.24	(44)	
	RadStd		0.16	0.33	0.34	0.42	0.49	0.39	0.31	0.25	0.38	0.26	0.24	0.13	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page